Skills Summary

Deciding Whether a Relation Is a Function

Use this reference while completing your worksheet or when studying.

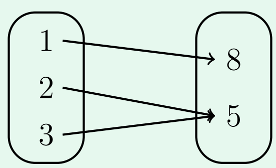
Definition. A relation is a **function** when every input has exactly one output.

- **The count test, in every representation:** count the pairs (arrows, rows, points), then count the *different* inputs. Equal counts $\Rightarrow$ function. Fewer inputs than pairs $\Rightarrow$ some input was used twice $\Rightarrow$ not a function.
- **Outputs may repeat. Inputs may not.** Two inputs sharing one output is fine; one input with two outputs is fatal.
- To *disprove* it, name the offending input and its two outputs. That one pair of values is the whole proof.

1. From a mapping diagram

Count the arrows, then count the dots on the left that have an arrow leaving them. A left-hand dot with two arrows is the only thing that can go wrong — two arrows arriving at the same right-hand dot is harmless.

Example: Is this mapping diagram a function?



Step 1. Do not look at the right-hand side first — only a repeated *input* can break the rule, so count arrows against left-hand dots.
Step 2. Arrows: three. Left-hand dots with an arrow leaving: three. So $\mathbf{3 = 3}$: no input is used twice, and it is a function. The two arrows landing on the same output do not matter.

Try it: A diagram has arrows $1 \to 4$, $6 \to 2$, $6 \to 9$, $8 \to 4$. Compare the arrow count with the count of different inputs.

check: $\mathbf{3 < 4}$ — not a function, since 6 has two outputs

2. From a set of pairs or a table

In $\{(x, y), \dots\}$ the *first* coordinate is the input; in a table it is the top row (or the left column). List the inputs, cross out repeats, and compare what is left with the number of pairs.

Example: Is $\{(2, 3),\ (4, 5),\ (6, 7),\ (2, 9),\ (8, 1)\}$ a function?
Step 1. The pairs are given, so the count test needs no drawing — just list the first coordinates and look for a repeat.
Step 2. First coordinates: $2, 4, 6, 2, 8$. Crossing out the repeated 2 leaves four different inputs among five pairs. So $\mathbf{4 < 5}$: the input 2 has outputs 3 and 9, so this is not a function.

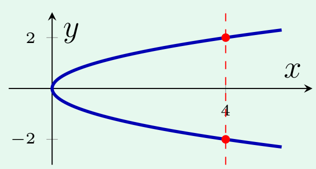
Try it: A table lists hours $1, 2, 3$ with costs $5, 5, 12$. Compare the number of rows with the number of different hours.

check: $\mathbf{3 = 3}$ — a function; the repeated cost is an output, so it is allowed

3. From a graph: the vertical line test

Slide a vertical line across the graph. Each position of the line is one input, and the number of crossings is the number of outputs it has. **More than one crossing anywhere ⇒ not a function.** One failing line is enough to decide it, so name the line and the two points it hits.

Example: The curve below is $x = y^2$. Is it a function?



Step 1. One failing vertical line settles it, so look for an input the curve reaches twice rather than checking the whole picture.
Step 2. The dashed line meets the curve at two marked points, so that one input has two outputs, against the one a function allows. So $\mathbf{1 < 2}$: the vertical line test fails and this is not a function.

Try it: For the parabola $y = x^2$, how many times does a vertical line meet the curve, against the one output a function allows?

check: $\mathbf{1 = 1}$ — exactly one crossing, so it is a function

(!) Watch out: being a function depends on which way the arrows point. "Student → locker" can be a function while "locker → student" is not, from the very same table. Always check which column you are calling the input.